

Dr. Saurabh Kumar

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EMPLOYMENT HISTORY:

Title/Position	University/Institution	Period
Postdoctoral Fellow (PDF)	MNNIT Allahabad, Prayagraj - under Visvesvaraya Ph.D. Scheme, Phase II, Digital India Corporation, MeitY, Govt. of India	September 2025 – August 2026
Senior Project Associate	Dept. of ECE, Madan Mohan Malaviya University of Technology (MMMUT), Gorakhpur - "Design of Low Power Memory Circuits", Chip to Startup (C2S) Programme, MeitY, GOI	August 2024 – September 2025

EDUCATION:

Ph.D. (Semiconductor Device Modeling) - August 2024

Madan Mohan Malaviya University of Technology, Gorakhpur, India

Title: Performance Investigation of Some Field Effect Transistors for Detection of Biomolecules.

M.Tech. (ECE) - December 2012, First Class with Distinction, CGPA 8.80

Amity University, Noida, India

Diploma in Embedded System and Design - August 2008, 85.3%

Center for Development of Advanced Computing (CDAC)

B.E. (Electronics and Communication Engineering) - September 2007

Visveswaraiiah Technological University, Belgaum, Karnataka, India

TEACHING INTEREST:

- CMOS Low Power Circuits Design
- Nano CMOS Technology
- Electronics (Analog and Digital) Fundamentals

FIELD OF INTEREST / RESEARCH INTERESTS:

- **VLSI Design:** Low-power digital, analog, and mixed-signal integrated circuit design.
- **AI Hardware:** Energy-efficient VLSI design for AI applications.
- **Memory Circuits:** Design and optimization of low-power SRAM architectures.
- **Semiconductor Devices and Biosensors:** Modeling, performance investigation, and optimization of MOS/FET devices for biomolecule detection and intelligent sensing applications.
- **Optoelectronic and Photovoltaic Devices:** Design, modeling, and performance optimization of photodetectors and thin-film solar cells.

SKILLS:

Analog & Mixed-Signal Circuit Design, Low-Power VLSI Design, Full-Custom IC Design Flow, ASIC Tape-out, CMOS Circuit Design, SRAM Design, PLL/VCO Design, Semiconductor Device Modeling, Circuit Performance Optimization, Cadence Virtuoso (Virtuoso, Spectre, ADE Explorer/Assembler, Layout XL), Verilog/VHDL, DRC/LVS/PEX, Monte Carlo and Corner Analysis, Project Management, Collaborative Research.

Journals:

1. S. Singh, **S. Kumar**, M. Deo, R. K. Chauhan, "Performance enhancement of methylammonium tin iodide perovskite photodetectors through device optimization," *Solar Energy*, Vol. 300, 2025. <https://doi.org/10.1016/j.solener.2025.113874> (IF: 6.6).
2. **S. Kumar**, R. K. Chauhan, "Synergistic impact of parametric variation and gate misalignment on gate-oxide stacked junctionless MOSFETs for biosensing applications," *AEU - International Journal of Electronics and Communications*, Vol. 201, 2025. <https://doi.org/10.1016/j.aeue.2025.156006> (IF: 3.2).
3. **S. Kumar**, R. K. Chauhan, M. Kumar, "Sensitivity Enhancement of Dual Gate FET Based Biosensor Using Modulated Dielectric for Covid Detection," *Silicon*, 14, 11453–11462 (2022). <https://doi.org/10.1007/s12633-022-01865-7> (IF: 3.3).
4. **S. Kumar**, R. K. Chauhan, "Biosensor sensitivity and steric hindrance: A comparison of InGaAs pocket and conventional TFET designs," *Micro and Nanostructures*, 182 (2023). <https://doi.org/10.1016/j.micrna.2023.207644> (IF: 3.0).
5. **S. Kumar**, R. K. Chauhan, "Tweaking the Performance of Dielectric Modulated Junctionless Double Gate Metal Oxide Field Effect Transistor-Based Label-Free Biosensor," *Journal of The Electrochemical Society*, 171(1), 017503, 2024. <https://doi.org/10.1149/1945-7111/ad1c19> (IF: 3.3).
6. S. Singh, **S. Kumar**, M. Deo, R. K. Chauhan, "Unveiling the potential of organometal halide perovskite materials in enhancing photodetector performance," *Optical and Quantum Electronics*, 55, 846 (2023). <https://doi.org/10.1007/s11082-023-05127-7> (IF:4.0).
7. **S. Kumar**, R. K. Chauhan, "Performance analysis of dielectric modulated underlap FD-SOI MOSFET for biomolecules detection," *Applied Physics A*, 128, 1104 (2022). <https://doi.org/10.1007/s00339-022-06260-y> (IF: 2.8).
8. **S. Kumar**, R. K. Chauhan, "Performance up-gradation of CIGS solar cell using Ag₂S quantum dot as buffer layer," *Journal of Materials Research*, 38, 2689–2700 (2023). <https://doi.org/10.1557/s43578-023-00992-0> (IF: 2.9).
9. S. Yadav, R. K. Chauhan, R. Mishra, **S. Kumar**, "Investigating Ag₂S quantum dot buffer layer in CZTSSe photovoltaics for enhanced performance," *Physica Scripta*, 98 (2023). <https://doi.org/10.1088/1402-4896/ad06fa> (IF: 2.6).
10. J. Singh, R. K. Chauhan, **S. Kumar**, "Design a double-gate perfectly aligned impact ionization MOSFET for RF and analog applications," *Physica Scripta*, 100, 7 (2025). <https://doi.org/10.1088/1402-4896/ade01e> (IF: 2.6).
11. S. Yadav, R. K. Chauhan, R. Mishra, **S. Kumar**, "Non-toxic materials for high efficiency thin-film solar cells: CZTSSe with In₂S₃ buffer layer," *Indian Journal of Physics* (2024). <https://doi.org/10.1007/s12648-024-03194-4> (IF: 1.7).
12. K. A. Verma, M. Kumar, **S. Kumar**, R. K. Chauhan, "Design of energy efficient domino logic circuit using lector technique," *International Journal of Electronics* (2022). <https://doi.org/10.1080/00207217.2022.2145500> (IF: 1.1).
13. B. Singh, **S. Kumar**, R. K. Chauhan, "Design of energy efficient VCO for PLL application," *Analog Integrated Circuits and Signal Processing*, 114, 31–40 (2023). <https://doi.org/10.1007/s10470-022-02122-y> (IF: 1.4).
14. S. Goel, **S. Kumar**, R. Tripathi, S. Bajpai, R. Soni, R. K. Chauhan, "Investigating the 20T Hybrid Full Adder Design for Low Power and High-Performance Computing," *IETE Journal of Research*, 1–8 (2024). <https://doi.org/10.1080/03772063.2024.2387292> (IF: 1.3).

15. **S. Kumar**, R. K. Chauhan, "A Novel Dielectric Modulated Misaligned Double-Gate Junctionless MOSFET as a Label-Free Biosensor," *Engineering Proceedings*, Vol. 35, 8 (2023). <https://doi.org/10.3390/IECB2023-14578>.
16. A. Singh, R. K. Chauhan, **S. Kumar**, Prince Kumar Singh, "Design and Performance Comparison of 6T, 8T and 11T SRAM Cells using TSMC and UMC 65nm CMOS Technology," *International Journal of Electronics* (2026). <https://doi.org/10.1080/00207217.2026.2664246> (IF: 1.1).

Published Book Chapters:

1. **S. Kumar**, R. K. Chauhan, Vimal Kumar Mishra, "A Perspective View for Detection of COVID-19 Causative Virus (SARS-CoV-2-) Using Dielectric Modulated Field Effect Transistors," Ch. 12 in *Opto-VLSI Devices and Circuits for Biomedical and Healthcare Applications*, Routledge & CRC Press, 2023. <https://doi.org/10.1201/9781003431138-12>.
2. V. Vishwakarma, **S. Kumar**, R. K. Chauhan, "Domino logic buffer circuit with reduced voltage swing for energy efficient application," Ch. 77 in *Emerging Trends in IoT and Computing Technologies*, Routledge & CRC Press, 2022. <https://doi.org/10.1201/9781003350057>.
3. B. Singh, **S. Kumar**, R. K. Chauhan, "Minimization of switching energy of domino logic circuits with T-NOT gate," Ch. 24 in *Emerging Trends in IoT and Computing Technologies*, Routledge & CRC Press, 2022. <https://doi.org/10.1201/9781003350057>.
4. S. Rai, M. D. Gupta, **S. Kumar**, R. K. Chauhan, "Improved Digital Performance of Modified Source-Drain FDSOI by Optimization of Buried Oxide Properties," in *VLSI, Microwave and Wireless Technologies, Lecture Notes in Electrical Engineering*, vol 877, Springer, 2022. https://doi.org/10.1007/978-981-19-0312-0_73.

Conference Papers:

1. V. Kumar, **S. Kumar**, V. Tiwari, S. K. Gupta and R. K. Nagaria, "Machine Learning-Assisted Optimization of Power Amplifier for Wireless Applications," 2026 1st International Conference on VLSI Circuits and Systems (ICVCS), Kolkata, India, 2026, pp. 1-6, <https://doi.org/10.1109/ICVCS69586.2026.1168316>.
2. **S. Kumar**, V. Tiwari, V. Kumar, S. K. Gupta and R. K. Nagaria, "Design and Analysis of an Enhanced Current-Starved Ring VCO Using NMOS-Only Stacked Delay Cells," 2026 IEEE Guwahati Subsection Conference (GCON), Guwahati, India, 2026, pp. 1-6, <https://doi.org/10.1109/GCON69192.2026.11648886>.
3. **S. Kumar**, R. K. Chauhan, M. Kumar, M. D. Gupta, "Impact of Different Technology Node on the Delay and Power Dissipation of 6T SRAM Cell," ICSC 2021, pp. 337-342. <https://doi.org/10.1109/ICSC53193.2021.9673326>
4. S. Singh, **S. Kumar**, R. K. Chauhan, "Optimizing Absorber Layer in Photodetectors with CH₃NH₃SnI₃ Organometal Halide Perovskite: A Simulation-Based Approach," ICICAT 2023, Gorakhpur, pp. 1-6. <https://doi.org/10.1109/ICICAT57735.2023.10263656>.
5. S. Mishra, **S. Kumar**, R. K. Chauhan, "Analysis of the Proposed Dynamic XNOR Circuit using Dual Keeper with Twist-Connected Transistors," DELCON 2022, pp. 1-4. <https://doi.org/10.1109/DELCON54057.2022.9753211>
6. S. P. Nayak, **S. Kumar**, R. K. Chauhan, "Design of Half Adder Domino Circuit Using Sleep and Twist-Connected Transistors with Dual Keeper," DELCON 2022, pp. 1-5. <https://doi.org/10.1109/DELCON54057.2022.9753457>
7. V. Kumar, N. K. Sharma, S. Srivastava, **S. Kumar**, P. Pandey, R. K. Chauhan, "Low Power SRAM cell Design Using MTCMOS and Stacking Techniques," ICICAT 2024, Gorakhpur, pp. 1665-1670. <https://doi.org/10.1109/ICICAT62666.2024.10923332>.

8. A. Anand, **S. Kumar**, R. K. Chauhan, "CNTFET-based SRAM cell Designing and Analysis by using Verilog-A code," ICICAT 2024, Gorakhpur, pp. 1558-1562. A code," ICICAT 2024, Gorakhpur, pp. 1558-1562. <https://doi.org/10.1109/ICICAT62666.2024.10922889>
9. A. Singh, R. K. Chauhan, **S. Kumar**, P. K. Singh, "Design and Simulation of Low Power Leakage Controlled SRAM Cell," ICSC 2025, Noida, pp. 797-802. <https://doi.org/10.1109/ICSC64553.2025.10968384>.

HONORS/AWARDS/ACHIEVEMENTS:

- **Visvesvaraya Postdoctoral Fellowship**, Ministry of Electronics and Information Technology (MeitY), Government of India, 2025.
- **Commendable Research Award**, Madan Mohan Malaviya University of Technology (MMMUT), Gorakhpur 26 August 2025. Awarded for excellence in research conducted during January - June 2025.
- Qualified the **UGC-NET examination** in Electronic Science for **Assistant Professor** eligibility in 2026.
- **Best Paper Award**, "Impact of Different Technology Nodes on the Delay and Power Dissipation of 6T SRAM Cell," ICVMWT-21, Madan Mohan Malaviya University of Technology (MMMUT), Gorakhpur - March 2021.
- **Best Poster Award**, "Impact of Variation of Underlap Region on Dielectric Modulated FET for Bio-Sensing," Madan Mohan Malaviya University of Technology (MMMUT), Gorakhpur - December 2021.
- **First Prize**, Poster Presentation, SPIN 2012, Amity University, Noida in 2012.
- M.Tech. with **First Division and Distinction**, Electronics and Communication Engineering, Amity University Uttar Pradesh in 2012.

FDP / SYMPOSIUM / IEP ATTENDED:

- **5-Day FDP** on Simulation, Modelling, and Application of Advanced Semiconductor Devices, Dept. of Electronics Engineering, Harcourt Butler Technical University, Kanpur, 05–09 July 2021.
- **5-Day Online FDP** "Application of AI in Water Resource Management (AAWRM-2022)," Rajkiya Engineering College Sonbhadra, 07–11 March 2022.
- **5-Day Instructor Enhancement Program (IEP)** under C2S - "ASIC Digital Design Using 180nm SCL," ChipIN Center, CDAC-Bangalore, 12–16 February 2024.

EXPERT LECTURES / RESOURCE PERSON:

- **Speaker**, "An overview of VLSI Technology: Device, Circuits and Simulation," FDP on Technology and Research in Electrical and Electronics Engineering, Shri Ramswaroop Memorial University, Lucknow, July 18–22, 2025 (online).
- **Expert lecture**, "An Overview of VLSI Technology: Devices, Circuits and LTspice Simulation," ESRVNE 2025, Dept. of Electronics Engineering, Rajkiya Engineering College Sonbhadra.
- **Provided VLSI training** to summer interns, VLSI Research Lab, Dept. of ECE, MMMUT, Gorakhpur, June–July 2024.

VOLUNTEER SERVICES:

Reviewer of Journals: Journal of Applied Electrochemistry (springer), Journal of Nanobiotechnology (springer), Plasmonics (springer), Engineering Research Express (springer), Circuit World (Emerald), Nano Select (Wiley), International Journal Of Electronics (Taylor & Francis), ACS Applied Materials & Interfaces (ACS).